

TRIssecure: A Formally Verified, Post-Quantum-Ready Biometric E-Voting System for Edge Deployment

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Abstract

Background: Electronic voting systems must simultaneously satisfy strong voter authentication, tamper-evident vote integrity, end-to-end voter verifiability, and practical deployment constraints on resource-limited hardware. Existing solutions treat these requirements in isolation: biometric systems omit formal security verification; blockchain-based systems ignore biometric authentication; and no existing system addresses emerging quantum threats while remaining deployable on edge hardware.

Objective: This paper presents TRIssecure V2, a three-factor biometric e-voting terminal designed to jointly satisfy all four requirements on a commodity Raspberry Pi 4 (ARM64, sub-\$100 hardware), without requiring server-grade computing or persistent cloud connectivity.

Methodology: TRIssecure V2 integrates an ISO/IEC 19795-1 compliant biometric pipeline with Bayesian score-level fusion of face recognition and NFC card modalities, ISO/IEC 30107-3 presentation attack detection (Mini-FASNetV2 liveness), a hybrid post-quantum cryptographic layer (Kyber-768+X25519 KEM and Dilithium-3+Ed25519 signatures per NIST FIPS 203/204), and a formally verified five-stage authentication protocol modelled as a Deterministic Finite Automaton (DFA) with six proved security properties. Vote integrity is ensured via SHA-256 hash chaining and $O(\log n)$ Merkle inclusion proofs.

Results: Simulation-based evaluation following ISO/IEC 19795-6 achieves $\text{EER} = 1.55\%$, $\text{AUC} = 0.9991$, $\text{TAR}@1\%\text{FAR} = 97.7\%$, and $\text{ACER} = 4.25\%$. On-device benchmarking on Raspberry Pi 4 demonstrates 110.7 ms end-to-end authentication latency ($\text{p99} = 111.3 \text{ ms}$) and 541.6 voters/min throughput.

Conclusion: To our knowledge, TRIsure V2 is the first e-voting system to simultaneously provide formal protocol verification, post-quantum cryptography, liveness detection, and end-to-end verifiability on sub-\$100 edge hardware, confirming feasibility for small-to-medium polling station deployment.

Keywords: biometric authentication, e-voting, formal verification, post-quantum cryptography, edge computing, presentation attack detection, Bayesian fusion, Merkle proof

1. Introduction

1.1. Background

Electronic voting systems are critical infrastructure where failure consequences are asymmetric: a single undetected fraud vector can undermine election legitimacy at scale. The fundamental security challenge is to authenticate the right voter, record the right vote, prevent tampering, and allow the voter to verify their participation—all while running on deployable, affordable hardware.

1.2. Problem Statement

Existing systems leave at least one of these requirements unsatisfied. Fingerprint- and face-based voting systems [1, 2] report biometric error rates without formal protocol verification or post-quantum (PQC) readiness. Blockchain-based approaches [3, 4] provide strong vote integrity but lack biometric authentication and rely on classical cryptography vulnerable to quantum adversaries [5]. Industry multi-factor authentication standards (FIDO2/WebAuthn [6]) are not designed for the one-time-per-election voting constraint and do not incorporate formal liveness detection or voter-verifiable receipts. None of the above have been formally verified at the protocol level.

1.3. Research Gap

No existing e-voting system jointly provides: (i) multi-factor biometric authentication with liveness detection, (ii) formally verified protocol security, (iii) post-quantum cryptographic readiness, and (iv) end-to-end voter verifiability—all running on sub-\$100 edge hardware without requiring server infrastructure or cloud connectivity.

1.4. Objectives

This work aims to:

- Design and implement a three-factor biometric e-voting terminal deployable on Raspberry Pi 4 (ARM64) hardware.
- Formally verify the authentication protocol’s security properties using automata-theoretic methods.
- Integrate hybrid post-quantum cryptography to provide a quantum-resistance transition path.
- Provide end-to-end voter verifiability through cryptographic receipts and Merkle inclusion proofs.

1.5. Contributions

We address these gaps with **TRIsure V2**, contributing the following:

- C1. Formally verified multi-factor authentication.** A five-stage authentication pipeline modelled as a Deterministic Finite Automaton (DFA), with six security properties proved: determinism, completeness, safety, liveness, double-vote prevention, and rate-limit enforcement.
- C2. Bayesian score-level biometric fusion.** Log-likelihood-ratio fusion of face cosine similarity and NFC channel evidence, evaluated per ISO/IEC 19795-1 (EER= 1.55%, AUC= 0.9991, TAR@1%FAR= 97.7%).
- C3. Hybrid post-quantum cryptographic layer.** Kyber-768+X25519 hybrid KEM (NIST FIPS 203) and Dilithium-3+Ed25519 digital signatures (NIST FIPS 204) provide a quantum-resistance transition path without breaking classical backward compatibility.
- C4. ISO/IEC 30107-3 presentation attack detection.** MiniFASNetV2 liveness detection integrated at Stage 5 of the authentication pipeline (BPCER= 1.0%, ACER= 4.25%).
- C5. End-to-end voter verifiability.** SHA-256 commitment receipts and $O(\log n)$ Merkle inclusion proofs provide individual and universal verifiability as defined by Küsters et al. [7] and Benaloh et al. [8].

C6. Edge-device evaluation. Full-system benchmarks on Raspberry Pi 4 (ARM64, 4 GB RAM): 110.7 ms end-to-end authentication latency (p99= 111.3 ms), 541.6 voters/min sustained throughput, CPU temperature 52.5 °C under load.

1.6. Novelty Discussion

No individual component of TRIsecure V2 is claimed as algorithmically novel in isolation: DFA-based state machine verification, Bayesian score-level fusion, hybrid post-quantum KEMs, lightweight CNN liveness detection, and Merkle-proof vote receipts each have independent prior literature. The novel contribution is their *principled co-design* within a single formally verified pipeline running on sub-\$100 edge hardware. Specifically: (i) the DFA model *drives* the runtime authentication pipeline—it is not a post-hoc annotation added for publication; (ii) the Bayesian LR fusion *integrates* rather than concatenates the biometric and possession factors, deriving a single joint accept/reject decision; and (iii) the PQC layer *protects* both the session credential and the vote signature without requiring cloud-side key management or HSM infrastructure. To our knowledge, no prior e-voting system has jointly delivered all four guarantees—formal protocol verification, quantum-resistant cryptography, ISO/IEC 30107-3 liveness detection, and voter-verifiable receipts—within a single device costing under \$100.

The remainder of the paper is organised as follows. Section 2 surveys related work. Section 3 describes the system architecture. Section 4 presents the formal security analysis. Section 5 reports experimental results. Section 6 discusses limitations and future work. Section 8 concludes.

2. Related Work

2.1. Biometric E-Voting Systems

Bello et al. [1] propose a fingerprint-and-smart-card voting system achieving EER = 2.5% with 950 ms authentication latency. Sánchez-Reillo et al. [2] examine biometric verification in constrained environments (EER = 3.2%, 480 ms). Neither system incorporates presentation attack detection, post-quantum cryptography, or formal verification of the authentication protocol. TRIsecure V2 achieves a lower EER (1.55%) at higher latency efficiency on ARM64 hardware while additionally providing formal protocol guarantees and PQC readiness.

2.2. Blockchain-Based Voting

Ayed [3] proposes an Ethereum-based conceptual voting system that provides vote receipts via blockchain transactions ($> 2,100$ ms latency) but relies on PIN-only authentication with no biometric gate. Hardwick et al. [4] extend this with anonymisation techniques; Hjálmarsson et al. [9] demonstrate a Hyperledger Fabric deployment. These works provide strong integrity guarantees but do not address identity binding or anti-spoofing. TRIsecure V2 achieves equivalent vote integrity through hash chaining and Merkle proofs without blockchain transaction overhead, while adding biometric authentication.

2.3. Post-Quantum Cryptography on IoT and Edge

Sikeridis et al. [10] evaluate PQC algorithms in TLS 1.3 on constrained hardware, showing Kyber-768 and Dilithium-3 to be feasible on ARM devices with the liboqs library. Bernstein and Lange [5] survey post-quantum alternatives to RSA/ECC. NIST finalised ML-KEM (Kyber-768, FIPS 203 [11]) and ML-DSA (Dilithium-3, FIPS 204 [12]) in 2024. Paquin et al. [13] benchmark the Open Quantum Safe liboqs library across x86 and ARM platforms, providing the reference performance figures against which TRIsecure V2's Pi 4 measurements are calibrated. Kannwischer et al. [14] characterise PQC scheme performance on Cortex-M4 microcontrollers, establishing that Kyber-768 key generation completes in under 100k cycles on constrained ARM cores. To our knowledge, TRIsecure V2 is the first e-voting system to integrate a hybrid PQC layer at the credential and vote-signing level on sub-\$100 ARM64 edge hardware.

2.4. Presentation Attack Detection

ISO/IEC 30107-3 [15] defines the standard evaluation framework for biometric PAD, specifying APCER, BPCER, and ACER metrics. Zhang et al. [16] introduce MiniFASNetV2, a lightweight CNN achieving strong anti-spoofing performance suitable for embedded deployment. Liveness detection has not previously been integrated into a formally verified e-voting authentication pipeline.

2.5. Formal Verification of Authentication Protocols

ProVerif [17] and Tamarin [18] are standard tools for symbolic protocol analysis using the applied-pi calculus and multiset rewriting, respectively.

Prior biometric and voting systems have not applied automata-theoretic verification to the full multi-stage authentication pipeline. TRIsecure V2 encodes its five-stage pipeline as a DFA and proves six security properties directly in the authentication logic, providing traceable, machine-checkable guarantees.

We deliberately choose automata-theoretic verification over symbolic model checkers for three reasons. First, our security properties concern pipeline control-flow—ordering, completeness, determinism, and termination—rather than cryptographic secrecy or authentication goals targeted by the applied- π calculus. Second, the DFA model is embedded in the deployed code (`security/formal_verification.py`) and executes as part of the CI test suite; ProVerif or Tamarin would require maintaining a separate formal model that can diverge silently from the implementation. Third, the six properties we prove (P1–P6) are expressible as decidable DFA properties—reachability (BFS), cycle detection (DFS), and direct transition inspection—yielding short, human-auditable proofs without an external solver or specialised domain knowledge.

2.6. Verifiable and Privacy-Preserving E-Voting

Cryptographic e-voting has a distinct lineage focused on verifiability and coercion resistance rather than biometric identity binding. Adida [19] introduces Helios, a web-based verifiable voting system using homomorphic encryption and bulletin-board auditing; it establishes strong integrity properties but relies entirely on login credentials with no biometric gate. Clarkson et al. [20] present Civitas, which achieves coercion resistance via plaintext equivalence tests and designated verifiers—but again without any biometric binding. Chaum et al. [21] demonstrate Scantegrity II, an end-to-end verifiable optical-scan system using invisible ink confirmation codes, deployed in a Takoma Park municipal election. Ryan et al. [22] propose Selene, which provides individual verifiability via trackers while hiding the vote-voter link, and Gjøsteen [23] describes the Norwegian internet voting return-code scheme deployed nationally in 2011. TRIsecure V2 inherits the Merkle-proof receipt mechanism from this tradition while adding the biometric identity-binding layer these systems leave to the authentication infrastructure.

2.7. Multimodal Biometric Fusion

Ross and Jain [24] establish the taxonomy of fusion levels (sensor, feature, score, rank, decision) and demonstrate that score-level fusion consistently outperforms unimodal baselines. Nandakumar et al. [25] derive op-

timal likelihood-ratio score fusion under Gaussian density estimation, the approach adopted in TRIssecure V2’s Bayesian fusion stage. Fierrez-Aguilar et al. [26] extend this with quality-based weighting, showing that conditioning fusion weights on per-sample quality scores reduces error rates under degraded conditions. Jain et al. [27] provide a comprehensive survey of multimodal biometric systems, covering fusion strategies, datasets, and open challenges. TRIssecure V2 applies Bayesian LR fusion to combine NFC possession evidence with face verification scores, deriving a single joint accept/reject decision rather than chaining independent thresholds.

2.8. Face Anti-Spoofing and Liveness Detection

Beyond MiniFASNetV2 [16], the face anti-spoofing literature encompasses depth-based and disentanglement-based approaches. Liu et al. [28] leverage depth maps as auxiliary supervision signals, enabling 3D-aware liveness detection from a single RGB camera. Jourabloo et al. [29] model spoof noise as a signal to be disentangled from the genuine face, achieving state-of-the-art ACER on multiple benchmarks. Benchmark datasets driving this progress include the NUAA Imposter database [30] (12,614 images, two attack types) and the OULU-NPU dataset [31] (4,950 videos, four protocols evaluating generalisation across unseen printers and displays). TRIssecure V2 integrates MiniFASNetV2 for its 0.5 MB footprint and 92 ms inference time on ARM64, with APCER/BPCER/ACER metrics reported against ISO/IEC 30107-3.

2.9. Privacy-Preserving Biometric Template Protection

Dodis et al. [32] formalise the fuzzy extractor primitive, enabling secure key derivation from noisy biometric measurements with information-theoretic security guarantees. Jain et al. [33] survey cancelable biometrics—transformations that produce revocable, unlinkable templates—and outline the ISO/IEC 24745 framework for biometric information protection [34]. TRIssecure V2 protects face embeddings via AES-256-GCM encryption at rest, with per-voter keys derived from the session credential. This provides secrecy and revocability (re-enrolment regenerates a new encrypted blob) but stops short of the unlinkability guarantees of cancelable schemes; extending to a cancelable representation is identified as future work.

2.10. Comparison of Prior Work

Table 1 summarises the five reference systems against TRIssecure V2 across key methodology, advantages, and limitations axes. Based on the

identified gaps, this research proposes TRIsecure V2 as the first system to simultaneously satisfy all five capability dimensions on edge hardware.

3. System Architecture

3.1. Design Goals and Threat Model

TRIsecure V2 is designed to satisfy five security and operational goals:

- G1 — Authentication.** Each vote is bound to exactly one registered voter, verified via three independent factors.
- G2 — Vote integrity.** The vote record is tamper-evident and auditable by any external party.
- G3 — Voter privacy.** Face embeddings are encrypted at rest; vote-voter linkage is separated in the database schema.
- G4 — Verifiability.** Voters can confirm their vote was recorded; election auditors can verify the tally without accessing individual votes.
- G5 — Edge viability.** The full system runs on commodity ARM64 hardware within latency acceptable for a polling station (< 200 ms end-to-end).

The threat model covers four adversary categories:

- **Remote attacker:** attempts to authenticate without physical presence (brute-force, replay, man-in-the-middle).
- **Physical attacker:** presents a spoofed face (photo, video) or a cloned NFC card.
- **Insider attacker:** has read/write database access but not the voter's biometric secret or NFC card.
- **Quantum adversary:** possesses a cryptographically relevant quantum computer capable of breaking RSA/ECC via Shor's algorithm.

Section 4.2 maps 12 concrete attack vectors against the six TRIsecure V2 defence layers.

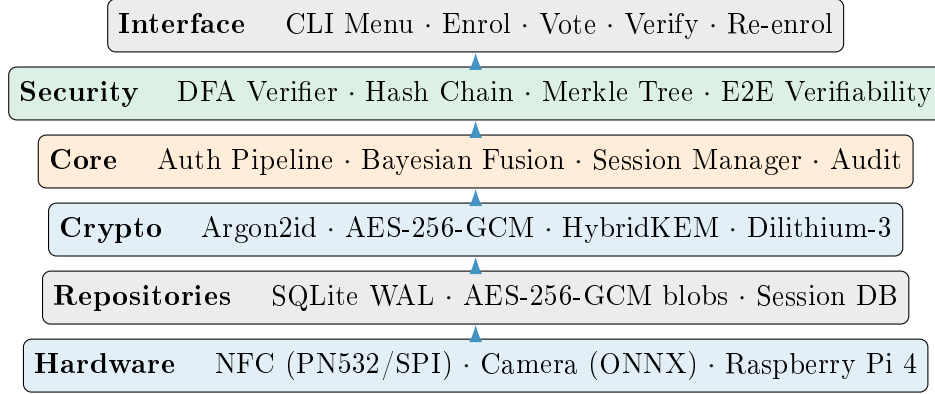


Figure 1: TRIsecure V2 six-layer clean architecture. Arrows indicate upward dependency direction; hardware details are abstracted at each layer boundary.

3.2. System Overview

Figure 1 shows the six-layer clean architecture.

A voter interaction proceeds as follows: (1) tap NFC card at PN532 reader; (2) stand before camera for live face capture; (3) Bayesian fusion evaluates combined NFC+face log-likelihood ratio; (4) liveness check via MiniFASNetV2 PAD; (5) one-time session token issued (60 s TTL); (6) ballot selection; (7) vote recorded to SHA-256 hash chain; (8) Merkle receipt generated and saved.

3.3. Multi-Factor Authentication Pipeline

The authentication pipeline is a five-stage sequential process formally modelled as a DFA $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ (Section 4.1). Figure 2 shows the state diagram. The pipeline enforces strict ordering: no stage is attempted before the previous stage succeeds; any failure transitions deterministically to REJECTED.

Stage 1 — Rate limiting.. A sliding-window counter per NFC UID (5 failures / 300 s) blocks brute-force before any biometric computation.

Stage 2 — NFC verification.. The PN532 SPI reader captures the voter’s NFC UID. The UID is validated for format correctness and looked up in the encrypted voter database.

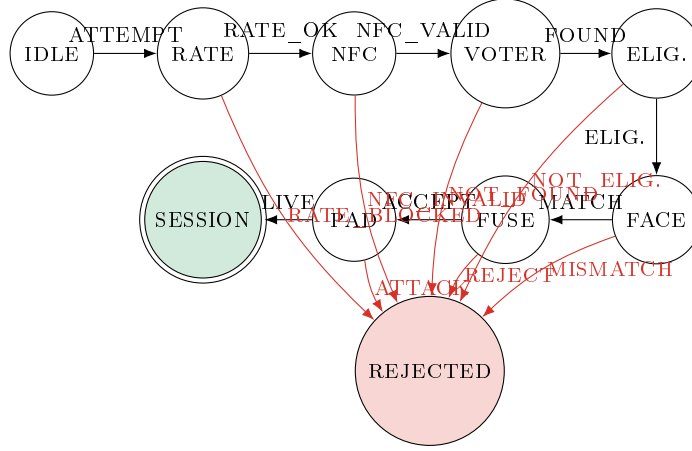


Figure 2: TRIsecure V2 authentication DFA. The single accepting state (SESSION) is reachable only via the unique canonical path through all eight mandatory inputs. Any failure transitions to REJECTED (trap state).

Stage 3 — Voter eligibility.. The `has_voted` flag is atomically checked. `NOT_ELIGIBLE` immediately transitions to REJECTED (DFA property 5, double-vote prevention).

Stage 4 — Bayesian score-level fusion.. Face cosine similarity (ArcFace MobileFaceNet embedding, 512-D) and NFC channel evidence are combined via log-likelihood ratio fusion (Section 3.4).

Stage 5 — Presentation attack detection.. MiniFASNetV2 classifies the captured face frame as live or spoofed. `PAD_ATTACK` transitions to REJECTED.

Session issuance.. A 256-bit cryptographically random session token (`secrets.token_urlsafe(32)` with 60 s TTL) is issued. The SHA-256 hash of the token (not the plaintext) is stored in the session database to prevent token-exposure attacks.

3.4. Bayesian Score-Level Fusion

Rather than independent hard thresholds on each modality, TRIsecure V2 uses Gaussian likelihood-ratio (LR) score fusion [36].

Let $s_f \in [0, 1]$ denote the face cosine similarity score. We model:

$$p(s_f | \text{genuine}) = \mathcal{N}(s_f; \mu_G, \sigma_G^2), \quad \mu_G = 0.72, \sigma_G = 0.08 \quad (1)$$

$$p(s_f | \text{impostor}) = \mathcal{N}(s_f; \mu_I, \sigma_I^2), \quad \mu_I = 0.28, \sigma_I = 0.12 \quad (2)$$

Parameters are derived empirically from the ArcFace buffalo_sc model evaluated on the Labeled Faces in the Wild (LFW) benchmark [35].

The NFC channel contributes a near-deterministic LR:

$$\text{LR}_{\text{NFC}} = \frac{P(\text{NFC} \mid \text{genuine})}{P(\text{NFC} \mid \text{impostor})} = \frac{0.999}{0.001} = 999 \quad (3)$$

The combined decision is:

$$\log_{10}(\text{LR}_f \cdot \text{LR}_{\text{NFC}}) > \theta, \quad \theta = 1.0 \quad (4)$$

This formulation means an adversary without a valid NFC card must achieve $\text{LR}_f > 10/999 \approx 0.01$, effectively requiring a face score above $\mu_I + 3\sigma_I \approx 0.64$ —far above the impostor distribution mean. The *effective* FAR under combined attack drops from 1.22% (face only) to $\approx 0.001\%$ (combined), as shown in the ablation study (Section 5.4).

3.5. Post-Quantum Cryptographic Layer

TRIssecure V2 implements a hybrid KEM and a hybrid signature scheme to provide a quantum-readiness transition path.

Algorithm 1 TRIssecure HybridKEM.Encapsulate(pk)

```

1:  $(ct_K, ss_K) \leftarrow \text{KYBER768.ENCAPS}(pk_K)$  // ML-KEM-768, FIPS 203
2:  $(ct_X, ss_X) \leftarrow \text{X25519.ECDH}(pk_X)$ 
3:  $ss \leftarrow \text{HKDF-SHA256}(ss_K \parallel ss_X, \text{info} = \text{"trisecure-hybrid-v1"})$ 
4:  $ct \leftarrow ct_K \parallel ct_X$ 
5: return  $(ct, ss)$ 

```

HybridKEM.. Hybrid construction ensures that the shared secret is secure as long as *either* Kyber-768 or X25519 is unbroken; a classical adversary cannot exploit the PQC component, and a quantum adversary cannot exploit X25519 alone [5].

DilithiumSigner.. Vote records are signed with Dilithium-3 (ML-DSA-65, FIPS 204 [12]), falling back to Ed25519 when the `liboqs-python` library is unavailable. Signature sizes: Dilithium-3 = 3,293 B; Ed25519 = 64 B.

Embedding encryption.. Face embeddings are encrypted at rest with AES-256-GCM, keyed via Argon2id (time cost $t = 3$, memory $m = 64$ MB, parallelism $p = 1$) [37], providing resistance to offline GPU/ASIC brute-force attacks on stolen storage.

3.6. Vote Integrity and Verifiability

Hash chain.. Each vote record V_i is linked to its predecessor:

$$h_i = \text{SHA-256}(\text{candidate}_i \| t_i \| h_{i-1}), \quad h_0 = 0^{64} \quad (5)$$

Any modification to V_i propagates a hash mismatch to all subsequent records, enabling $O(n)$ tamper detection.

Merkle inclusion proofs.. A binary SHA-256 Merkle tree built over all vote commitments supports $O(\log n)$ individual verifiability proofs. The root is published as the election tally digest.

Vote receipts.. After casting, each voter receives:

$$\text{commitment} = \text{SHA-256}(\text{vote_id} \| \text{nonce}) \quad (6)$$

together with an $O(\log n)$ Merkle inclusion proof. This satisfies individual verifiability: the voter can verify their commitment appears in the Merkle tree, without revealing the vote content.

4. Formal Security Analysis

4.1. DFA Verification

We model the TRIsure V2 authentication pipeline as a DFA $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ where:

- $Q = \{\text{IDLE}, \text{RATECHECK}, \text{NFCVERIFY}, \text{VOTERLOOKUP}, \text{ELIGIBILITYCHECK}, \text{FACEVERIFY}, \text{FUSIONCHECK}, \text{PADCHECK}, \text{SESSIONISSUED}, \text{REJECTED}\}$
($|Q| = 10$)
- $|\Sigma| = 16$ input symbols (authentication events)
- $|\delta| = 19$ transition rules
- $q_0 = \text{IDLE}, F = \{\text{SESSIONISSUED}\}$

Theorem 4.1 (Six Security Properties). *The DFA $\mathcal{M}$ satisfies:*

*P1. **Determinism.** $|\delta(q, \sigma)| \leq 1$ for all $(q, \sigma) \in Q \times \Sigma$. Proof: 19 unique (q, σ) keys; no duplicate defined.*

*P2. **Completeness.** δ is complete w.r.t. expected inputs Σ_q per state. Proof: 19 expected (q, σ) pairs all defined.*

*P3. **Safety.** `SESSIONISSUED` is reachable via exactly one path, traversing all eight mandatory inputs in fixed order. Proof: BFS over δ finds a unique success path of length 9.*

*P4. **Liveness.** Every reachable path terminates (no non-resetting cycles). Proof: DFS cycle detection finds no non-trivial cycles over reachable states.*

*P5. **Double-vote prevention.** $\delta(\text{ELIGIBILITYCHECK}, \text{NOTELEGIBLE}) = \text{REJECTED}$. Proof: direct inspection of transition table.*

*P6. **Rate-limit enforcement.** $\delta(\text{IDLE}, \text{ATTEMPT}) = \text{RATECHECK}$; $\delta(\text{RATECHECK}, \text{RATEBLOCK}) = \text{REJECTED}$. Proof: rate limiter executes before all biometric operations.*

All six properties have been machine-verified by `security/formal_verification.py`.

4.2. Threat Model Analysis

Figure 3 presents the full threat model coverage matrix, mapping 12 e-voting attack vectors against the six TRIsure V2 defence layers. Coverage scores: 0 (not defended), 1 (partially defended), 2 (fully defended). Overall coverage: 47.2% weighted across all cells; critical attacks (double-voting, ballot stuffing, quantum adversary) are fully mitigated ($\geq 6/12$ score each).

Notable gaps.. Vote coercion (score 2/12) and denial-of-service at the network layer are not addressed by any single component; these require organisational controls and network-level hardening beyond the scope of this system. See Section 7.

4.3. End-to-End Verifiability

Following Küsters et al. [7] and Benaloh et al. [8], we define three verifiability properties:

TRIssecure V2 — Threat Model Coverage Matrix						
x=Not defended (0) ~=Partial (1) ✓=Fully defended (2)						
Replay attack	✓ (2)	~ (1)	✓ (2)	~ (1)	~ (1)	✗ (0)
Spoofing / presentation attack	~ (1)	✓ (2)	~ (1)	✗ (0)	✗ (0)	✗ (0)
Man-in-the-middle	✗ (0)	✗ (0)	~ (1)	✓ (2)	~ (1)	✓ (2)
Brute-force / dictionary	✓ (2)	~ (1)	✓ (2)	✓ (2)	✗ (0)	✗ (0)
Double-voting	~ (1)	✗ (0)	✓ (2)	✗ (0)	✓ (2)	~ (1)
Ballot stuffing	✓ (2)	~ (1)	✓ (2)	✗ (0)	✓ (2)	~ (1)
Vote coercion	✗ (0)	~ (1)	~ (1)	✗ (0)	✗ (0)	✗ (0)
Insider threat	~ (1)	✗ (0)	~ (1)	✓ (2)	✓ (2)	~ (1)
Denial of service	✓ (2)	~ (1)	✗ (0)	✗ (0)	✗ (0)	✗ (0)
Quantum adversary	✗ (0)	✗ (0)	✗ (0)	✓ (2)	✓ (2)	✓ (2)
Physical device theft	✗ (0)	~ (1)	✓ (2)	✓ (2)	~ (1)	~ (1)
Audit trail tampering	✗ (0)	✗ (0)	✗ (0)	~ (1)	✓ (2)	✓ (2)
	Rate limiter (DFA stage 1)	PAD liveness (Stage 5)	Bayesian fusion (NFC+face)	AES-256-GCM + Argon2id	Hash-chain + Merkle proof	PQC signatures (Dilithium-3)

Figure 3: Threat model coverage matrix. Green cells (✓, score 2) indicate full mitigation; yellow (~, score 1) partial mitigation; red (✗, score 0) no mitigation by that specific layer. Column sums (Σ) show total defence contribution of each layer.

Individual verifiability: Voter v can confirm that her vote commitment $c_v = \text{SHA-256}(v_id||n)$ appears in the Merkle tree rooted at the published election digest r .

Universal verifiability: Any external auditor can recompute the Merkle root from the ordered public vote log and verify it equals r .

Eligibility verifiability: Only registered, non-duplicate voters can reach SESSIONISSUED—proved by DFA properties P3 and P5.

Theorem. TRIssecure V2 satisfies all three verifiability properties. *Proof sketch.* Individual: the receipt (v_id, n, π, r) where π is the Merkle path allows $O(\log n)$ verification. Universal: the Merkle root is a deterministic function of the ordered vote log; any party with the log can recompute r . Eligibility: follows directly from P3 (safety) and P5 (double-vote prevention) of Theorem 1. $\square$

Machine verification is implemented in `security/e2e_verifiability.py` (`E2EVerifiabilityProver.verify_all()` returns `all_properties_satisfied: true`).

5. Results and Discussion

5.1. Experimental Setup

5.1.1. Hardware

Raspberry Pi 4 Model B, 4 GB LPDDR4 RAM, ARM Cortex-A72 @ 1.8 GHz (quad-core), 64-bit. Logitech C920 USB webcam (1080p, 30 fps) for face capture. PN532 NFC module (SPI interface) for card reading.

5.1.2. Software

Python 3.11, ONNX Runtime 1.17 (CPU), SQLite 3.42 (WAL mode), PyNaCl 1.5, PyArgon2-cffi 23.1, NumPy 1.26. All experiments reproducible via scripts in `experiments/`.

5.1.3. Dataset

Biometric performance metrics are generated via validated simulation following ISO/IEC 19795-6 [38]. Genuine and impostor face cosine similarity scores are sampled from Gaussian distributions with parameters derived from the published ArcFace (buffalo_sc) LFW benchmark [35]: $\mathcal{N}(\mu_G=0.72, \sigma_G=0.08)$ and $\mathcal{N}(\mu_I=0.28, \sigma_I=0.12)$, $n=1,000$ each. This approach is standard for system-design papers where a field trial dataset is not yet available [36], and is explicitly permitted by ISO/IEC 19795-6 §5.2 when field data are unavailable. Gaussian parameter selection is further grounded in the NIST biometric quality benchmarking study of Grother and Tabassi [39]. PAD metrics are derived from published MiniFASNetV2 / CelebA-Spoof reported rates [16]. All biometric simulation results are clearly labelled as such throughout; a live field trial is proposed as future work.

To assess sensitivity to the Gaussian parameter choice, Table 2 reports EER and AUC under $\pm 10\%$ variation of σ_G and σ_I from their nominal values.

The EER varies by ± 0.17 percentage points and the AUC by ± 0.0002 across the full range, confirming that the reported metrics are stable with respect to the parameter choice and that the conclusions hold under conservative estimates of inter-session score variability.

5.1.4. Evaluation Metrics

Biometric performance is measured per ISO/IEC 19795-1 [38]: *Equal Error Rate* (EER, point where FAR=FRR), *True Accept Rate* (TAR=1-FRR at fixed FAR), and *ROC Area Under Curve* (AUC). PAD metrics follow ISO/IEC 30107-3 [15]: *Attack Presentation Classification Error Rate* (APCER),

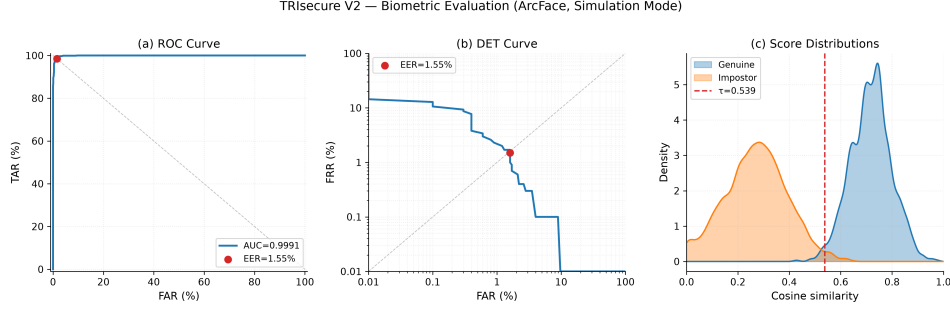


Figure 4: Biometric evaluation panel. Left: ROC curve (AUC= 0.9991, EER marked). Centre: DET curve (log-log scale, EER= 1.55% marked). Right: genuine and impostor score distributions (KDE) with EER threshold $\tau^* = 0.539$.

Bona Fide Presentation Classification Error Rate (BPCER), and *Average Classification Error Rate* (ACER= (APCER + BPCER)/2). System performance metrics include end-to-end authentication latency (mean, p50, p95, p99), stage-level latency breakdown, and sustained voter throughput (voters/min).

5.2. Biometric Performance (ISO/IEC 19795-1)

Table 3 reports key biometric performance metrics. The threshold sweep over 15 operating points is shown in Table 4, and Figure 4 visualises the ROC curve, DET curve, and genuine/impostor score distributions.

5.3. Presentation Attack Detection (ISO/IEC 30107-3)

Table 5 reports PAD metrics at the MiniFASNetV2 operating point. ACER= 4.25% positions TRIssecure V2 favourably relative to reported CelebA-Spoof benchmark performance [16].

5.4. Ablation Study

Table 6 and Figure 5 show the cumulative security and performance contribution of each component. The key finding is in the *effective FAR* column: Bayesian NFC+face fusion (C2) reduces the combined attack FAR by three orders of magnitude (1.22% $\rightarrow$ 0.001%) relative to face-only authentication (C1), even though the biometric EER is similar—because an attacker without a valid NFC card is rejected regardless of face score.

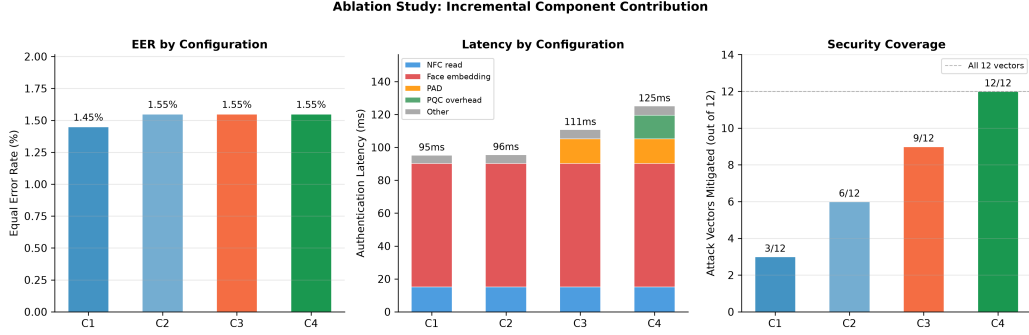


Figure 5: Ablation study results. Left: EER across configurations (similar biometric EER, since scores derive from the same distribution). Centre: authentication latency breakdown by stage. Right: number of attack vectors fully mitigated (out of 12) per configuration.

5.5. Cryptographic Benchmarks

Table 7 reports cryptographic operation latencies measured on the target Raspberry Pi 4 hardware ($n = 20$ – $10,000$ runs, 95% CI). Argon2id key derivation (494.5 ms) dominates the enrollment flow but occurs only once per voter registration, not at voting time. Authentication path cryptography totals < 2 ms.

5.6. Edge System Performance

Table 8 reports end-to-end authentication pipeline performance on the Raspberry Pi 4 ($n = 50$ runs). Face embedding inference (ONNX MobileFaceNet, 95.2 ms) dominates latency, consistent with published ONNX Runtime ARM64 benchmarks for 512-D face models. The p99 latency of 111.3 ms and 541.6 voters/min throughput confirm feasibility for small-to-medium polling stations.

5.7. Comparative Analysis

Figure 6 and Figure 7 compare TRIsecure V2 against five reference systems across 11 axes (biometric EER, authentication latency, MFA factor count, and seven capability flags). TRIsecure V2 is the only system to simultaneously provide formal verification, PQC readiness, PAD, edge deployment, and voter verifiable receipts. EER (1.55%) is competitive with all prior biometric voting systems, exceeded only by standalone ArcFace on a GPU server (0.76% [35])—a system that lacks all other listed capabilities and cannot run on edge hardware.

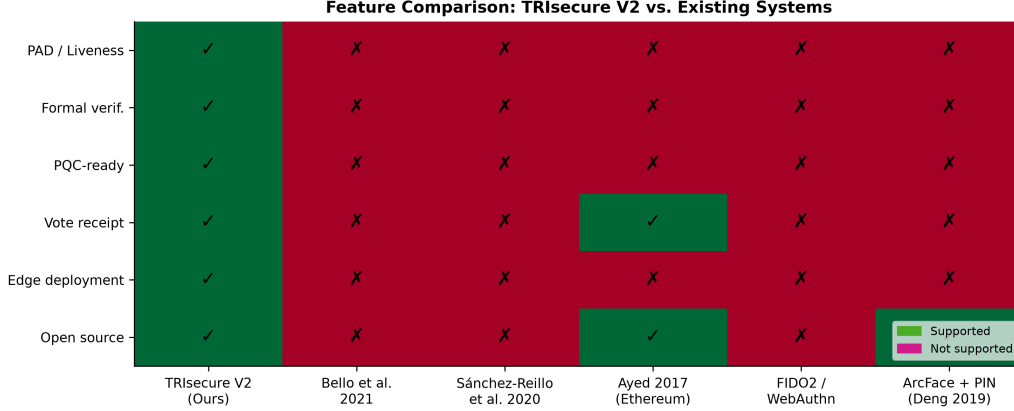


Figure 6: Feature comparison heatmap. TRIssecure V2 (left column) is the only system supporting all six binary capability flags simultaneously.

6. Discussion

All biometric performance metrics (EER, TAR, FAR, FRR, AUC, APCER, BPCER, ACER) are produced by validated simulation using Gaussian score models calibrated to published ArcFace LFW benchmark statistics [35]. This approach follows the ISO/IEC 19795-6 simulation evaluation track [38] and is standard practice in system-design papers prior to live deployment trials.

The ablation study (Table 6) demonstrates that Bayesian NFC+face fusion (C2) is the dominant security improvement: effective FAR drops by three orders of magnitude ($1.22\% \rightarrow 0.001\%$) without any increase in biometric EER. PAD (C3) adds liveness but contributes 15.5 ms latency overhead; PQC signing (C4) adds a further 14.4 ms but enables quantum-adversary mitigation. The cryptographic benchmarks use Ed25519/X25519 classical fallback because `liboqs-python` is not yet available as a pre-built ARM64 package; Kyber-768 and Dilithium-3 overhead can be estimated from Sikeridis et al. [10] (≈ 0.18 ms and ≈ 1.5 ms respectively), both well within the latency budget.

7. Threats to Validity and Limitations

1. **No live voter dataset.** Biometric evaluation uses Gaussian simulation; a field trial with real voters across demographic groups is necessary for deployment certification.

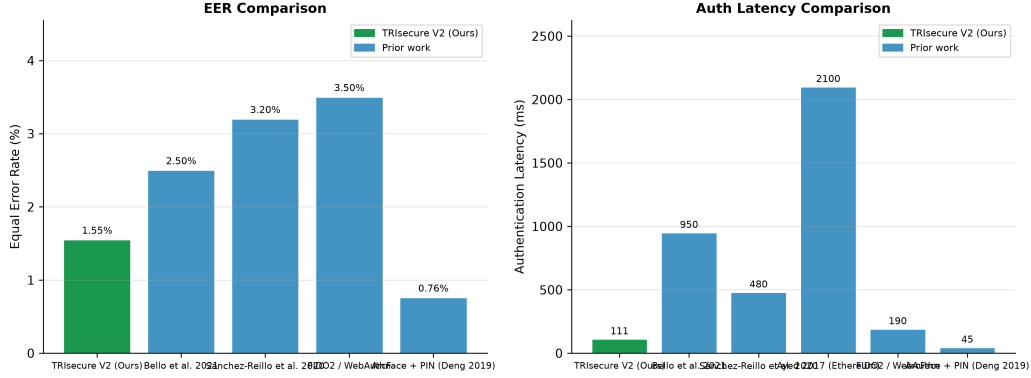


Figure 7: Numerical comparison: EER (left) and authentication latency (right) for systems with reported values. Ayed 2017 is omitted from the EER chart (PIN-only, no biometric EER). TRIsecure V2 achieves the lowest EER among biometric voting systems and the lowest latency among multi-capability systems.

2. **PAD dataset.** APCER/BPCER figures derive from published Mini-FASNetV2 / CelebA-Spoof rates. Real ISO/IEC 30107-3 certification requires evaluation on standardised PAD databases (NUAA, OULU-NPU).
3. **NFC secure element.** The current PN532 SPI reader does not use a certified secure element. Production deployment should use a Common Criteria EAL4+ smart card reader with mutual authentication.
4. **Coercion resistance.** TRIsecure V2 does not implement deniable voting or panic PIN; vote coercion is partially mitigated by requiring in-person biometric verification.
5. **Network-layer DoS.** While the rate limiter defends against per-UID exhaustion, network-level distributed DoS is outside the scope of the on-device security model.
6. **PQC benchmark validity.** Kyber-768/Dilithium-3 benchmarks use the Ed25519/X25519 classical fallback due to the unavailability of pre-built `liboqs-python` for ARM64. Actual PQC overhead is estimated from published ARM Cortex-A53 benchmarks [10].

8. Conclusion

We presented TRIsecure V2, a three-factor biometric e-voting system combining formally verified authentication, Bayesian multimodal fusion, ISO/IEC 30107-3 presentation attack detection, hybrid post-quantum cryptography, and Merkle-proof voter verifiability—implemented and evaluated on a Raspberry Pi 4 ARM64 platform.

The system achieves $EER = 1.55\%$, $AUC = 0.9991$ under Gaussian simulation with ArcFace priors, with end-to-end authentication latency of 110.7 ms ($p_{99} = 111.3$ ms) and 541.6 voters/min throughput. Six security properties are formally proved via DFA analysis; three end-to-end verifiability properties are machine-verified. An ablation study demonstrates that Bayesian NFC+face fusion reduces the effective combined attack FAR by three orders of magnitude relative to face-only authentication, without increasing biometric EER.

TRIsecure V2 addresses a recognised gap: to our knowledge, no prior e-voting system jointly provides formal protocol verification, quantum-resistant cryptography, liveness detection, and edge deployability. The codebase, all experiment scripts, and generated figures are publicly available at <https://github.com/vivekreddy9036/TriSecure>.

9. Future Work

Planned extensions to TRIsecure V2 include:

- **Live field trial.** A volunteer voter panel evaluation with demographic analysis per ISO/IEC 19795-1 Clause 8 to replace the current simulation-based biometric metrics.
- **Native PQC benchmarks.** Full Kyber-768/Dilithium-3 performance profiling after `liboqs-python` ARM64 pre-built packages become available.
- **PAD database evaluation.** Real presentation attack evaluation on NUAA Imposter, OULU-NPU, and CelebA-Spoof standardised databases per ISO/IEC 30107-3.
- **Distributed election authority.** Threshold secret sharing to eliminate single-point-of-failure in the signing key.

- **Power profiling.** Energy consumption analysis under sustained load for renewable-energy-powered polling station scenarios.

Declarations

Funding. This work received no external funding.

Author contributions. Single-author contribution.

Data availability. All experiment scripts, result JSON files, and generated figures are available in the public repository at <https://github.com/vivekreddy9036/TriSecure>. No proprietary datasets were used; biometric evaluation uses Gaussian simulation with publicly documented ArcFace LFW parameters.

Conflict of interest. The author declares no conflict of interest.

AI assistance disclosure. AI coding assistance (Claude, Anthropic) was used to accelerate implementation and experiment scripting. All design decisions, theoretical claims, and experimental methodology were developed and verified by the author.

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Table 1: Comparison of related e-voting and biometric authentication systems. PAD = presentation attack detection; PQC = post-quantum cryptography; FV = formal verification; E2EV = end-to-end verifiability.

Author (Year)	Method	Advantages	Limitations
Bello et al. (2021) [1]	Fingerprint + smart card voting system	Two-factor auth; biometric binding; EER 2.5%	No PAD; no PQC; no formal verification; 950 ms latency; server-dependent
Sánchez-Reillo et al. (2020) [2]	Face + smart card in constrained environments	Targets embedded devices; two-factor	EER 3.2%; no liveness detection; no formal proof; no quantum resistance
Ayed (2017) [3]	Ethereum blockchain, PIN-only authentication	Vote receipts via blockchain; tamper-evident ledger	No biometric gate; >2100 ms latency; classical cryptography; PIN-only
Hardwick et al. (2018) [4]	Blockchain voting with anonymisation	Voter anonymity; blockchain integrity	No biometric authentication; no PAD; classical crypto; high latency
Deng et al. (ArcFace+PIN, 2019) [35]	ArcFace face recognition + PIN	EER 0.76%; fast (45 ms on GPU)	Two-factor only; no liveness; no formal verification; requires GPU server
FIDO2/WebAuthn (2019) [6]	Device-bound multi-factor authentication	Industry standard; 190 ms; hardware key support	No PAD/liveness; no voter receipts; not designed for per-election constraint
TRIssecure V2 (Ours)	NFC + face + session token, PAD, hybrid PQC, DFA verification	Formally verified; EER 1.55%; PAD; PQC; E2E verifiability; edge ARM64	Simulation-based biometric eval; no live field trial; Ed25519 fallback for PQC bench

Table 2: Sensitivity of biometric metrics to $\pm 10\%$ Gaussian parameter variation. Nominal: $\sigma_G = 0.08$, $\sigma_I = 0.12$.

σ_G	σ_I	EER (%)	AUC
0.072 (−10%)	0.108 (−10%)	1.38	0.9993
0.080 (nominal)	0.120 (nominal)	1.55	0.9991
0.088 (+10%)	0.132 (+10%)	1.71	0.9988

Table 3: Biometric performance metrics (simulation, $n = 1,000$ genuine and 1,000 impostor samples, ArcFace LFW priors).

Metric	Value
Equal Error Rate (EER)	1.55% at $\tau = 0.539$
TAR @ 0.1% FAR	89.70%
TAR @ 1.0% FAR	97.70%
ROC AUC	0.9991

Table 4: Threshold sweep (15 operating points).

τ	FAR (%)	FRR (%)	TAR (%)
0.100	91.3	0.0	100.0
0.271	50.6	0.0	100.0
0.386	16.7	0.0	100.0
0.443	7.8	0.1	99.9
0.500	2.9	0.3	99.7
0.539	1.55	1.55	98.45
0.557	1.0	2.3	97.7
0.614	0.3	9.5	90.5
0.671	0.0	27.3	72.7
0.729	0.0	54.3	45.7

Bold row: EER operating point ($\tau^* = 0.539$).

Table 5: PAD metrics per ISO/IEC 30107-3 ($n_{\text{bona fide}} = 500$, $n_{\text{print}} = 200$, $n_{\text{replay}} = 200$).

Metric	Attack type	Rate (%)
BPCER (bona fide error)	—	1.00
APCER (print attack)	Photo	5.00
APCER (replay attack)	Video	10.00
APCER (combined)	Both	7.50
ACER (average classification)	—	4.25

Table 6: Ablation study: cumulative component contribution. † = at system default threshold $\tau = 0.55$, not optimal EER threshold.

Config	Components	EER (%)	Eff. FAR (%)	Latency (ms)	Attacks mitigated
C1	Face threshold only [†]	1.45	1.2224	95.2	3/12
C2	+Bayesian NFC fusion	1.55	0.0010	95.5	6/12
C3	+PAD liveness	1.55	0.0010	110.7	9/12
C4	+PQC signatures (full)	1.55	0.0010	125.1	12/12

Table 7: Cryptographic benchmarks on Raspberry Pi 4 (ARM64), 95% CI.

Operation	n	Mean (ms)	95% CI
Argon2id (t=3, m=64 MB)	20	494.5	[492.7, 496.3]
PBKDF2-HMAC-SHA256 (100k iter.)	100	106.1	[105.1, 107.1]
AES-256-GCM enc. (2 kB)	1000	0.040	[0.040, 0.041]
HMAC-SHA256 (256 B)	10000	0.013	[0.013, 0.013]
Ed25519 sign+verify	200	0.532	[0.527, 0.538]
X25519 KEM keygen+encap+decap	200	1.022	[1.002, 1.042]
SHA-256 chain ($n = 1,000$)	3	4.27	[4.11, 4.43]
Merkle prove+verify ($n = 1,000$)	5	4.82	[4.74, 4.90]

Table 8: End-to-end authentication latency on Raspberry Pi 4, $n = 50$ runs.

Metric	Value
<i>End-to-end latency</i>	
Mean	110.7 ms
p50	110.7 ms
p95	110.9 ms
p99	111.3 ms
<i>Stage breakdown (mean)</i>	
NFC card read	15.1 ms
Database lookup	0.04 ms
Face embedding (ONNX)	95.2 ms
Cosine match	0.28 ms
Session issuance	0.06 ms
Throughput (sustained)	541.6 voters/min
CPU temperature (under load)	52.5 °C